

ASSIGNMENT

4. TAKE ONE DOMAIN AND BUILD BUSINESS UNDERSTANDING

1. Introduction

Business understanding is the first and most critical phase in any analytics, data science, or AI project. It focuses on understanding the business objectives, problems, constraints, stakeholders, and success criteria before moving to technical implementation.



In this report, we take the E-Commerce domain as a case study and build a detailed business understanding framework covering objectives, challenges, KPIs, stakeholders, data needs, risks, and strategic recommendations.

Overview of the E-Commerce Domain

E-commerce refers to buying and selling goods or services over the internet. It includes:

- B2C (Business to Consumer)

- B2B (Business to Business)
- C2C (Consumer to Consumer)
- D2C (Direct to Consumer)

Major global examples include:

- Amazon
- Flipkart
- Alibaba Group
- Myntra

The e-commerce industry is driven by digital payments, smartphones, logistics networks, and data-driven personalization.

Business Context

➤ Industry Environment

The e-commerce market is highly competitive, price-sensitive, and customer-driven. Companies compete based on:

- Pricing
- Delivery speed
- Product variety
- Customer service
- Personalization
- Discounts and offers

➤ **Target Market**

Example: Mid-sized Indian online retail platform targeting:

- Age group: 18–40 years
- Urban & semi-urban customers
- Mobile-first users

- Price-sensitive shoppers

Problem Statement

The company is facing:

1. High cart abandonment rate
2. Low customer retention
3. Increasing customer acquisition cost
4. High return rate
5. Inefficient inventory management

The goal is to build business understanding to design data-driven solutions.

Business Objectives

Primary Objectives

1. Increase sales revenue by 20% in one year
2. Improve customer retention by 15%
3. Reduce cart abandonment rate by 10%
4. Optimize inventory to reduce holding cost

Secondary Objectives

- Improve customer experience
- Increase repeat purchases
- Improve marketing ROI
- Reduce operational inefficiencies

Stakeholder Analysis

Stakeholder	Role	Interest
CEO	Strategic decisions	Revenue growth
Marketing Team	Campaigns	Conversion rate
Operations Team	Logistics	Delivery efficiency
IT Team	Platform stability	System performance
Customers	Buyers	Better experience

Understanding stakeholder expectations ensures alignment between business and analytics teams.

Key Business Metrics (KPIs)

➤ Revenue Metrics

- Gross Merchandise Value (GMV)
- Net Revenue
- Average Order Value (AOV)

➤ Customer Metrics

- Customer Acquisition Cost (CAC)
- Customer Lifetime Value (CLV)
- Retention Rate
- Churn Rate

➤ Operational Metrics

- Inventory Turnover Ratio
- Order Fulfillment Time

- Return Rate
- Marketing Metrics
- Click Through Rate (CTR)
- Conversion Rate
- Cost per Click (CPC)
- Return on Ad Spend (ROAS)

Business Constraints



- 8.1 Financial Constraints
- Limited marketing budget
- High logistics cost
- 8.2 Technical Constraints
- Legacy systems
- Limited real-time analytics
- 8.3 Regulatory Constraints
- Data protection laws
- GST compliance (India)

➤ 8.4 Operational Constraints

- Delivery delays
- Supply chain disruptions

Data Requirements

To solve business problems, the following data is required:

➤ 9.1 Customer Data

- Demographics
- Browsing history
- Purchase history
- Payment method

➤ 9.2 Transaction Data

- Order ID
- Product ID
- Quantity
- Price
- Timestamp

➤ 9.3 Product Data

- Category
- Brand
- Cost
- Stock levels

➤ 9.4 Marketing Data

- Campaign performance
- Ad impressions

- Clicks

10. Business Understanding for Key Problems

➤ 10.1 Cart Abandonment

Possible Causes:

- High shipping charges
- Complicated checkout process
- Lack of payment options
- Slow website

Business Strategy:

- Offer free shipping above threshold
- Simplify checkout
- Provide multiple payment gateways
- Improve UI/UX

➤ 10.2 Customer Retention

Causes of Low Retention:

- Poor customer service
- Late deliveries
- No loyalty program

Strategy:

- Introduce loyalty rewards
- Personalized product recommendations
- Fast delivery options

➤ 10.3 Inventory Optimization

Problems:

- Overstocking
- Stockouts

Strategy:

- Demand forecasting
- ABC analysis
- Seasonal trend analysis

SWOT Analysis

Strengths

- Wide product range
- Digital payment integration

Weaknesses

- High competition
- Thin profit margins

Opportunities

- Rural market expansion
- AI-based personalization

Threats

- Regulatory changes
- Price wars
- Logistics disruptions

Risk Assessment

Risk	Impact	Mitigation
Data breach	High	Strong cybersecurity

Risk	Impact	Mitigation
Platform downtime	Medium	Cloud infrastructure
Negative reviews	High	Improve service quality

Competitive Analysis

Companies like Amazon and Flipkart use:

- AI-based recommendations
- Fast logistics networks
- Customer-centric policies
- Subscription models

To compete, mid-sized platforms must focus on niche markets and personalization.

Business Success Criteria

The project will be successful if:

- Revenue increases by 20%
- Cart abandonment reduces by 10%
- Customer retention improves by 15%
- Inventory cost reduces by 8–10%

Strategic Recommendations

1. Implement AI-driven recommendation system
2. Use predictive analytics for demand forecasting

3. Introduce subscription-based loyalty model
4. Improve website performance
5. Optimize supply chain using data analytics



Future Scope

- AI chatbots for customer service
- Dynamic pricing models
- Real-time personalization
- Voice commerce integration
- Expansion into international markets

Conclusion

Business understanding is the foundation of any successful analytics or AI initiative. In the E-Commerce domain, understanding customer behavior, operational inefficiencies, and revenue drivers is critical.

By clearly defining:

- Business objectives

- KPIs
- Stakeholders
- Constraints
- Data requirements
- Risks

The organization can build effective data-driven strategies to increase revenue, improve customer satisfaction, and sustain competitive advantage.

A structured business understanding approach ensures that technical solutions align with real business problems and deliver measurable value.